Decoding Biomedical Time Series into Natural Language

Using Hybrid Quantum-Classical Transformers

Phase01: Signal Preprocessing

PFE Presentation | 2026

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Agenda Overview

01 Dataset Architecture

02 Preprocessing Pipeline

03 Feature Extraction Pipeline

04 Discussion Questions

Dataset Architecture:

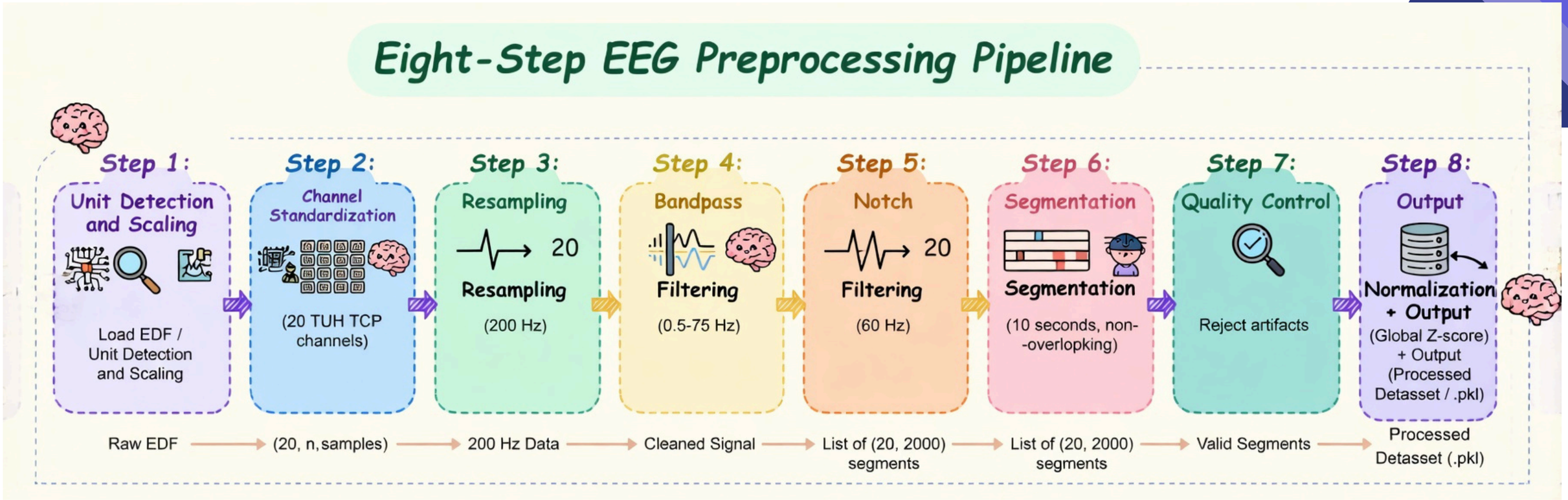
- **The TUH EEG Corpus is organized into multiple task-specific sub-corpora:**

Dataset	Task	Files	Hours	Labels
TUAB	Abnormal Detection	20,000	55.56	Normal/Abnormal
TUAR	Artifact Classification	19,999	55.55	Artifact types
TUEP	Epilepsy Detection	20,390	56.64	Epilepsy/Normal
TUEV	Event Classification	10,540	29.28	6 event types
TUSL	Slowing Detection	6,063	16.84	Slowing regions
TUSZ	Seizure Detection	20,112	55.87	Seizure/Normal
Total	**6 tasks**	**97,104**	**269.74**	**Multiple**

TUEV ,6 event types : bckg ,artf,spsw,gped,pled,eyem

TUSZ ,9 event types : bckg ,fnsz,gnsz,spsz,cpsz,absz,mysz,tnsz,tcsz

Preprocessing Pipeline:



Step 1: Unit Detection and Scaling:

Problem Statement

- MNE-Python ALWAYS returns EEG data in volts (SI units), regardless of

EDF header specifications. Quality control thresholds defined in microvolts caused all segments to fail.

Step 2: Channel Standardization:

20-Channel TUH TCP Montage

- We standardize all recordings to a 20-channel TCP (Temporal-Central-Parasagittal) bipolar montage. Note that the full TCP montage has 22 channels, but TUH monopolar recordings lack A1 and A2 ear reference electrodes, preventing formation of A1-T3 and T4-A2 derivations.

Monopolar → Bipolar Conversion?

1.Monopolar (Reference) Recording:

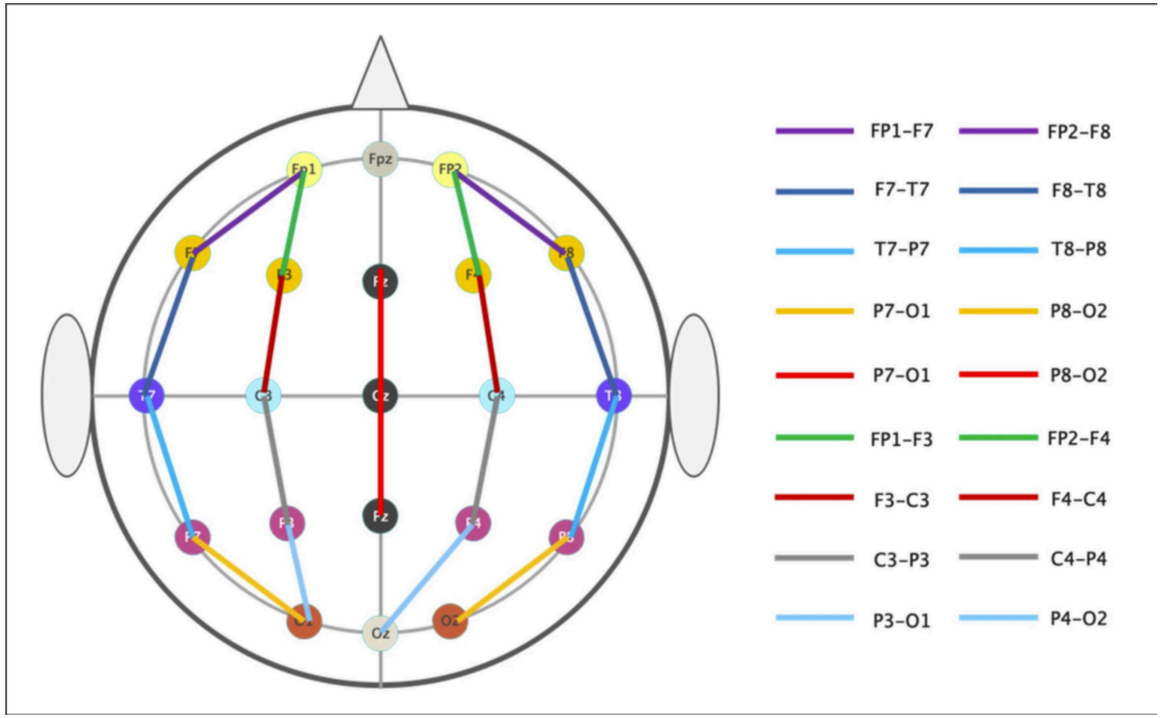
- Each electrode measured against a common reference (e.g., "EEG FP1-REF")
- Reference point: Cz, linked ears, or average of all electrodes
- Problem: Reference contamination affects ALL channels simultaneously

2.Bipolar Recording:

- Difference between two adjacent electrodes (e.g., "FP1-F7")
- No external reference needed
- Advantage: Common noise cancels out (subtraction)

Step 2: Channel Standardization:

20-Channel TUH TCP Montage



Chain	Bipolar Derivations
Left temporal	FP1-F7, F7-T3, T3-T5, T5-O1 (4)
Right temporal	FP2-F8, F8-T4, T4-T6, T6-O2 (4)
Central	T3-C3, C3-CZ, CZ-C4, C4-T4 (4)
Left parasagittal	FP1-F3, F3-C3, C3-P3, P3-O1 (4)
Right parasagittal	FP2-F4, F4-C4, C4-P4, P4-O2 (4)
Total	**20 channels**

Step 3: Resampling:

- All signals are resampled to 200 Hz using polyphase FIR with anti-aliasing filter. This provides adequate coverage for clinical EEG frequencies (up to 70 Hz) while reducing computational load.

Implementation

$$x_{\text{resampled}}[n] = \sum_k x_{\text{original}}[k] \cdot h\left[n \cdot \frac{f_{\text{orig}}}{f_{\text{target}}} - k\right]$$

Where:

- Xresampled[n] = Output sample at index n (200 Hz)
- Xoriginal[k] = Input samples (250-1000 Hz)
- h[n] = Windowed sinc function (anti-aliasing filter)
- forig = Original sampling rate (250-1000 Hz)
- ftarget = Target sampling rate (200 Hz)

Step 4: Bandpass Filtering:

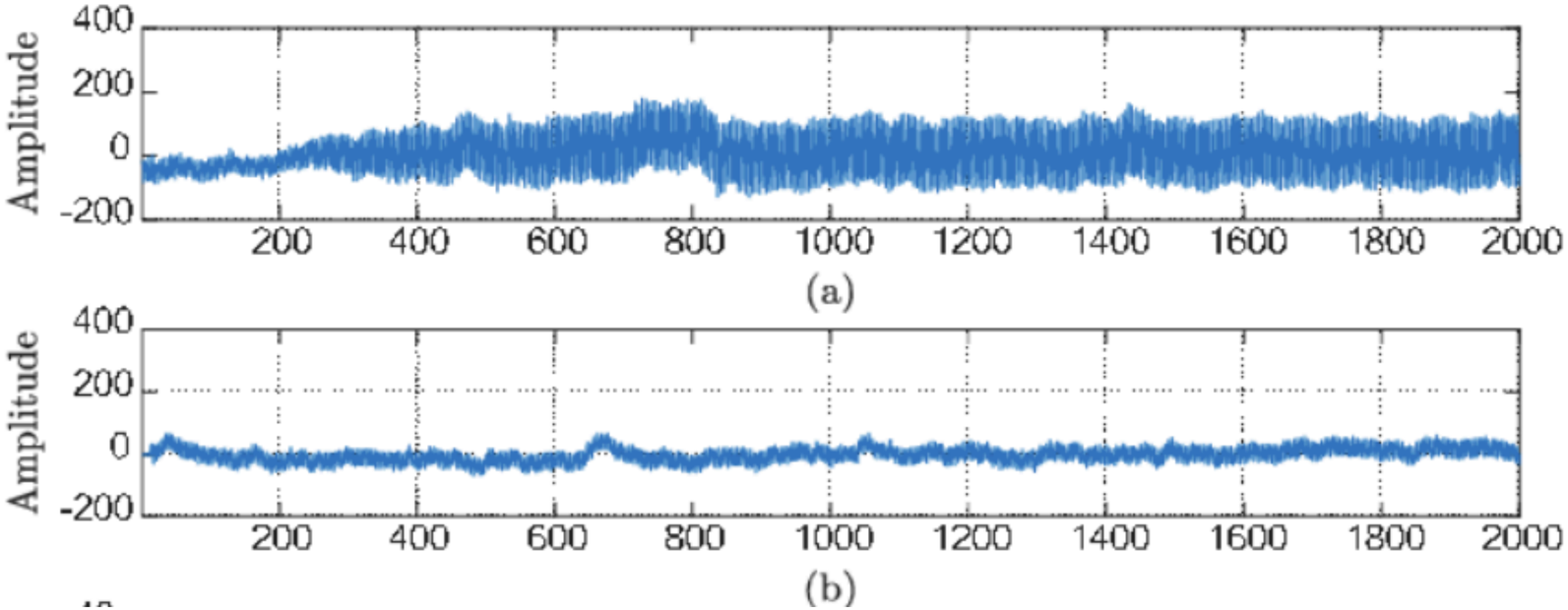
Filter Specifications

Parameter	Value
Lower cutoff frequency	0.5 Hz
Upper cutoff frequency	75 Hz
Filter type	Finite Impulse Response (FIR)
Phase response	Zero-phase (symmetric)

Step 5: Notch Filtering:

Powerline Noise Removal

- Removes 60 Hz powerline interference (North America) using spectrum-fit notch filter.



- (a) the raw EEG signal,
- (b) notch (60Hz) filtered EEG signal .

Step 6: Segmentation:

Segmentation Configuration

- **Segment Duration:** 10 seconds
- **Overlap:** 0% (non-overlapping windows)
- **Samples per Segment:** 2,000 (10 s $\times$ 200 Hz)
- **Output Dimensions:** (20 channels, 2000 samples)

Step 7: Quality Control:

Check	Criterion	Threshold
Amplitude	95th percentile of $\lvert X \rvert$	$< 500\ \mu V$
Median variance	Median of channel STDs	$> 0.5\ \mu V$
Flat channels	Proportion with STD $< 0.1\ \mu V$	$< 50\%$
Signal range	Peak-to-peak amplitude	$> 1.0\ \mu V$
		None allowed

Step 8: Normalization:

Global Z-Score Normalization

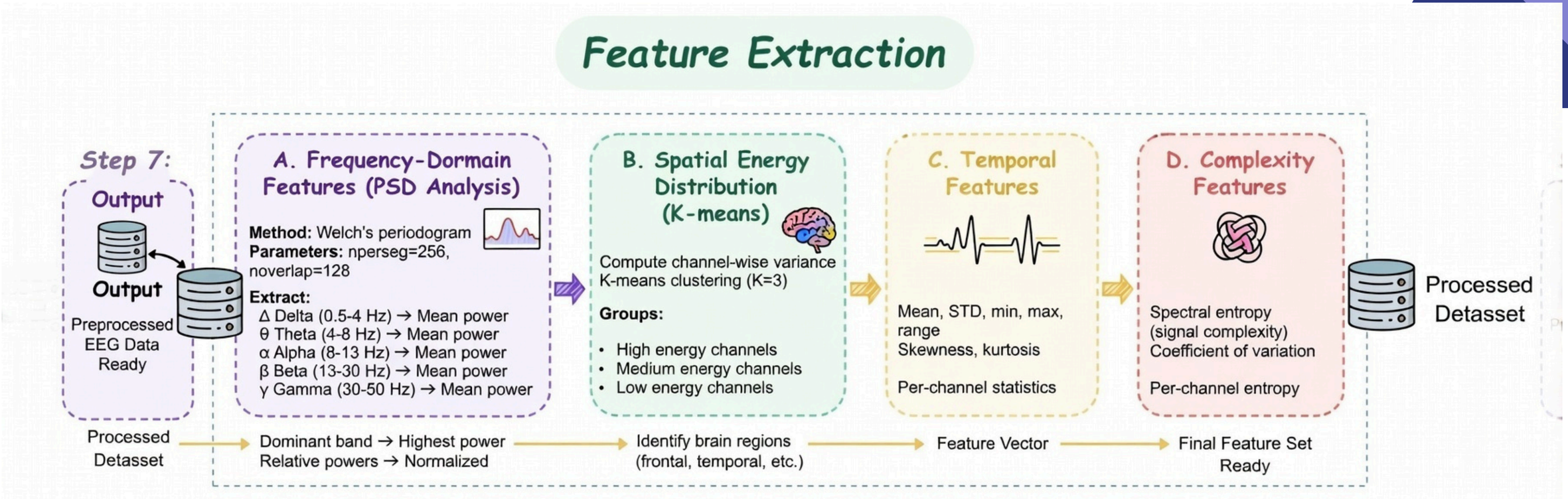
- We apply global z-score normalization to standardize data distribution:

$$X_{\text{norm}} = \frac{X - \mu}{\sigma + \varepsilon}$$

where:

- $\mu = \text{mean}(X)$ across all channels and samples
- $\sigma = \text{std}(X)$ across all channels and samples
- $\varepsilon = 10^{-8}$ for numerical stability

Feature Extraction Pipeline:



Step 1: Frequency-Domain Features

Power Spectral Density Estimation

- We use Welch's method for robust PSD estimation:

$$P_{xx}(f) = \frac{1}{K} \sum_{i=0}^{K-1} |\mathrm{FFT}(x_i \cdot w)|^2$$

- Where:
- Pxx(f): Power at frequency f
- K: number of segments
- xi: i-th segment of signal
- w: window function
- FFT: Fast Fourier Transform converts signal → frequency domain
- | . | ^2: power (energy)

Step 1: Frequency-Domain Features

Frequency Band Power Extraction

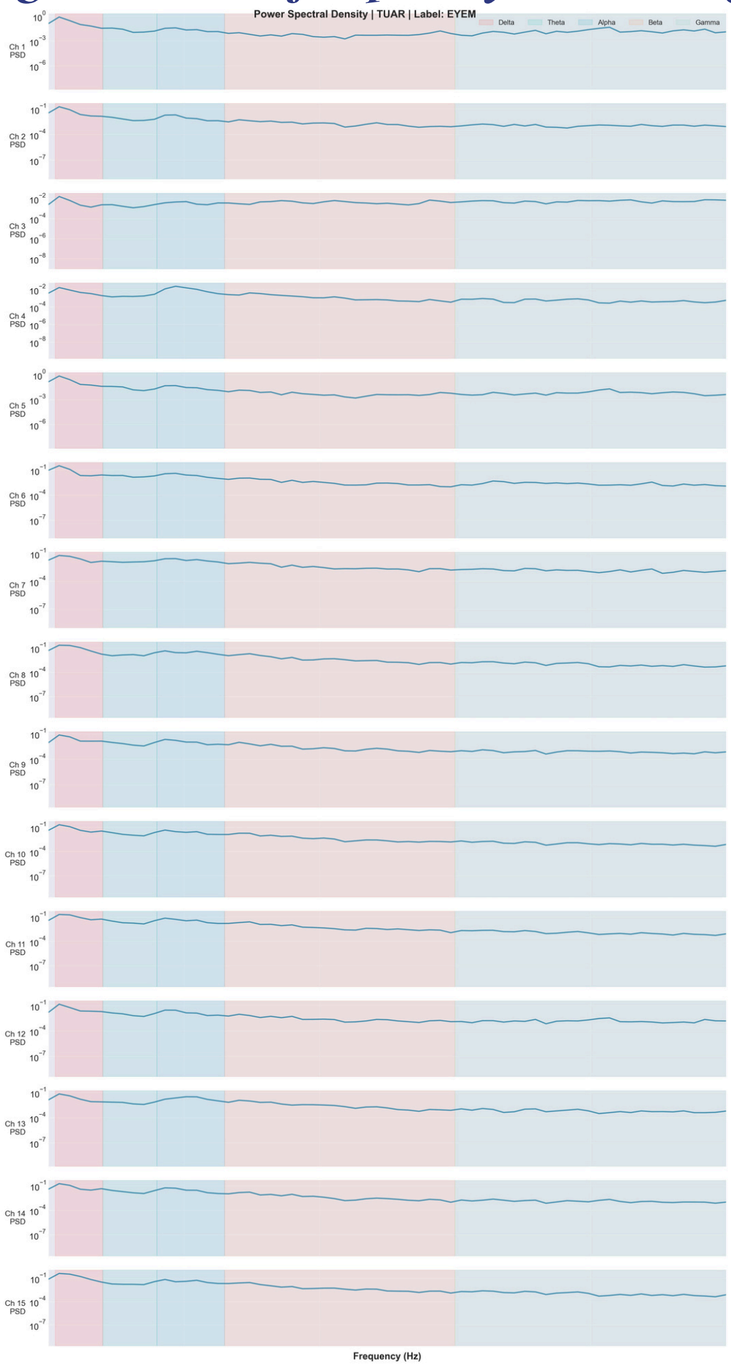
- For each standard frequency band Bi , we compute:

$$P_{B_i} = \int_{f_{\text{low}}^{(i)}}^{f_{\text{high}}^{(i)}} P_{xx}(f)\,df \approx \sum_{k:f_k \in B_i} P_{xx}(f_k) \cdot \Delta f$$

- Where:
- PBi: total power in band B
- Pxx(f): PSD (power at frequency f)
- flow,fhigh: band limits
- Δf: frequency resolution

Feature	Description
`band_powers`	Absolute power in each band: $\{P_\delta, P_\theta, P_\alpha, P_\beta, P_\gamma\}$
`relative_powers`	Normalized powers: $P_i / \sum_j P_j$
`dominant_band`	Band with maximum power: $\arg\max_i P_i$
`total_power`	Sum across all bands: $\sum_i P_i$
`band_powers_per_channel`	Per-channel band powers (20×5 matrix)
`psd_freqs`	Frequency bins (for visualization)
`psd_mean`	Mean PSD across channels
`psd_std`	Standard deviation of PSD across channels

Example PSD for a typical EEG segment with frequency bands highlighted.Dominant band: delta



Step 2: Spatial Features

- We cluster channels based on signal power to identify spatial patterns:

Algorithm 3 Spatial Energy Clustering

Require: EEG segment $X \in \mathbb{R}^{20\times 2000}$
Ensure: Spatial feature dictionary

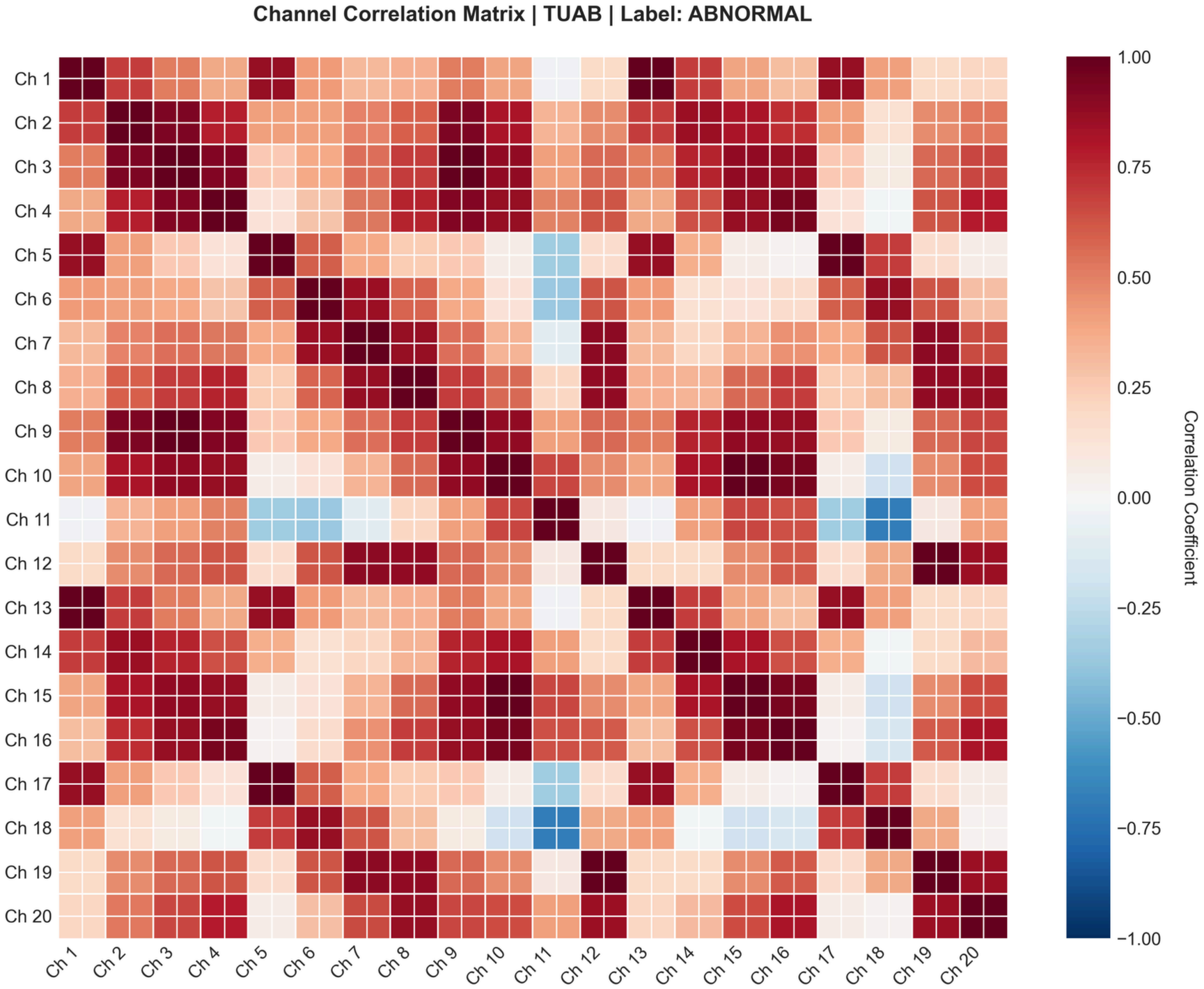
```
Compute channel powers: p_i = var(X_{i,:}) for i = 1, ..., 20
Find peak channel: i_peak = arg max_i p_i
Apply K-means with K = 3 on {p_1, p_2, ..., p_20}
Sort clusters by mean power: Low < Medium < High
Map each channel to energy level: {low, medium, high}
Identify brain regions for high-energy channels
Determine lateralization (left/right/bilateral)
return {peak, high_channels, medium_channels, low_channels, regions}
```

Example:
Focal seizure: Peak channel at T3, high-energy cluster includes F7-T3 and T3-T5, region = left temporal, lateralization = left.

Table 5.2: Brain Region Identification

Region	Channels
Left temporal	FP1-F7, F7-T3, T3-T5, T5-O1 (4)
Right temporal	FP2-F8, F8-T4, T4-T6, T6-O2 (4)
Central	T3-C3, C3-CZ, CZ-C4, C4-T4 (4)
Left parasagittal	FP1-F3, F3-C3, C3-P3, P3-O1 (4)
Right parasagittal	FP2-F4, F4-C4, C4-P4, P4-O2 (4)

$$\text{Lateralization} = \begin{cases} \text{Left} & \text{if } |C_{\text{left}}| > 1.5 \cdot |C_{\text{right}}| \\ \text{Right} & \text{if } |C_{\text{right}}| > 1.5 \cdot |C_{\text{left}}| \\ \text{Bilateral} & \text{otherwise} \end{cases}$$


Channel Correlation Matrix | TUAB | Label: ABNORMAL
Ch 1
Ch 2
Ch 3
Ch 4
Ch 5
Ch 6
Ch 7
Ch 8
Ch 9
Ch 10
Ch 11
Ch 12
Ch 13
Ch 14
Ch 15
Ch 16
Ch 17
Ch 18
Ch 19
Ch 20
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Correlation Coefficient

Step 3: Temporal Features

- Statistical moments characterize the amplitude distribution:

Table 5.3: Extracted temporal features

Feature	Formula	Interpretation
Mean	$\mu = \frac{1}{N}\sum x_i$	Central tendency
Std Dev	$\sigma = \sqrt{\frac{1}{N}\sum(x_i - \mu)^2}$	Signal variability
Range	$r = \max(x) - \min(x)$	Peak-to-peak amplitude
Skewness	$\gamma_1 = \frac{1}{N\sigma^3}\sum(x_i - \mu)^3$	Asymmetry
Kurtosis	$\gamma_2 = \frac{1}{N\sigma^4}\sum(x_i - \mu)^4$	Tailedness

Thank You

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